

00 A CONCEPT, EXPLAINED —

Essential complexity.

On the model that outlives the tools – from business architecture to code.

Robert Blust · Software Engineer & Architect

From business architecture to code – end-to-end traceability.

Four areas shaped my career: software development, methodology, IT architecture and business architecture. I know all four.

The goal is always end-to-end continuity – from the business domain to implementation. Only the path there was long expensive and hard.



A model captures a problem in its essential complexity – no more.

Capture the essential, leave out the unimportant. That is the real craft.

| My standard for every model, for about 15 years.



BEFORE

UML · SysML · your own DSL

Powerful – but hard, specialized, expensive. Few can read it.

Same clarity – far less friction.

TODAY

AI understands natural language.

With the right meta-model: the facts as Markdown – cheap to create and maintain, readable for everyone.

The Mental Model.

A structured knowledge base – I call it the Mental Model.

Domains & terms

Booking · Finance · Door Access · Task

How we work

Roles · teams · processes with gates

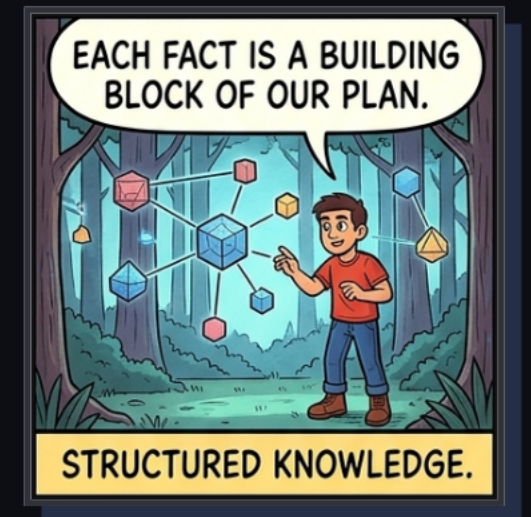
Why this way

Architecture decisions · rules

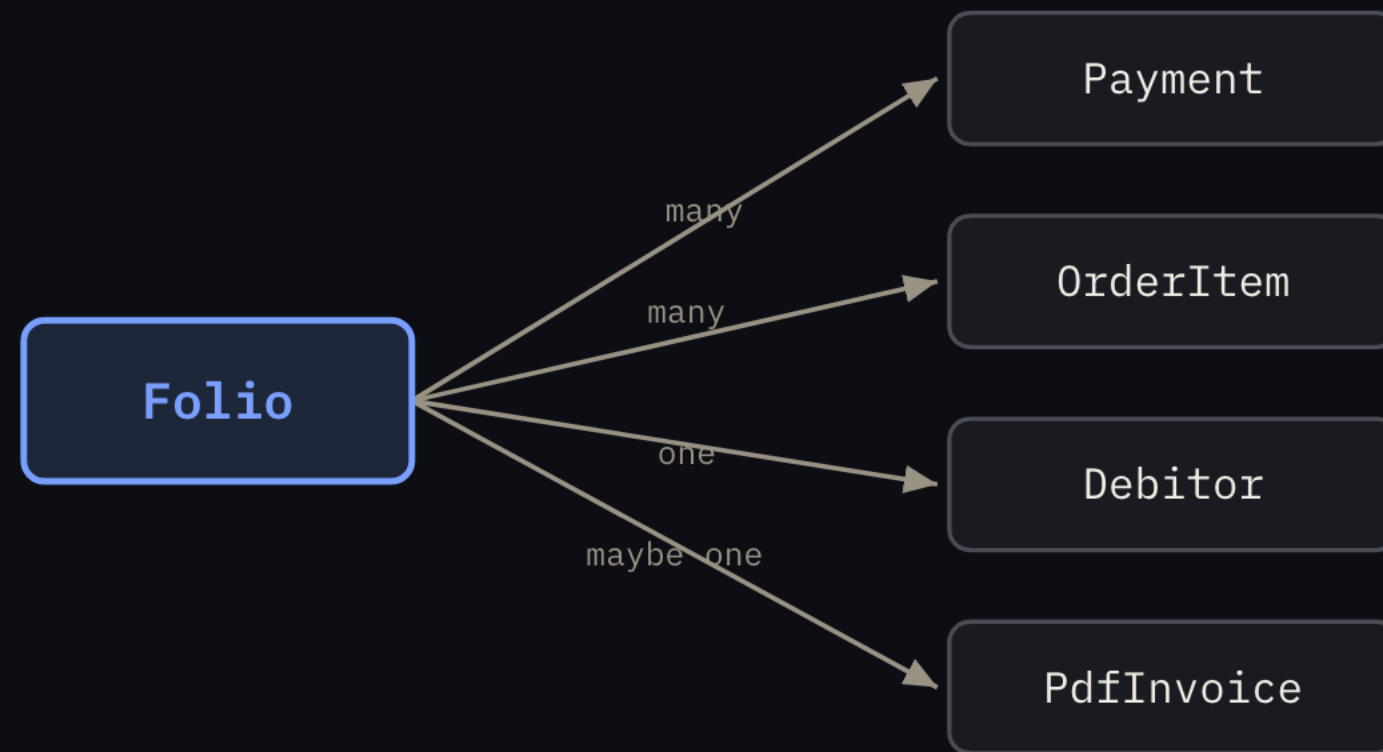
How we measure

KPIs

Facts as Markdown · structured by a meta-model · **machine-readable**



05 AN EXAMPLE - FINANCE IN A HOTEL SYSTEM



Folio

Financial account of a reservation – bundles all line items of a stay.

OrderItem

A single line item on the folio.

Payment

Payment that settles order items fully or partly.

Debitor

Who owes the bill – person or company.

PdfInvoice

The finalized invoice at check-out.

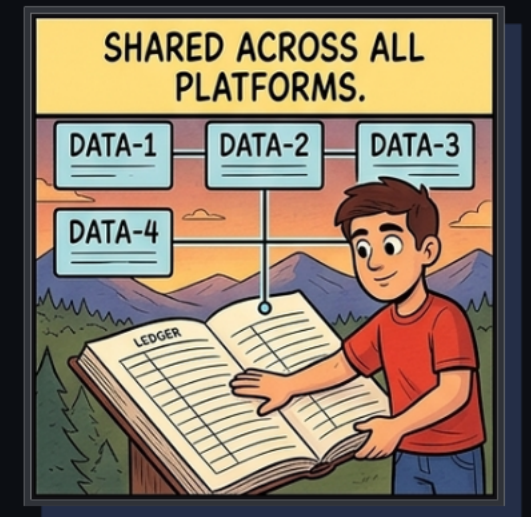
Real model from LIKE MAGIC – **Markdown + Mermaid**

Every term in one sentence – **no more, no less.**

One model that everything can be measured against.

- ✓ **Requirements** use only terms from the **glossary** – one language, no synonyms.
- ✓ **Implementation** validated against **glossary** + **ADRs**.
- ✓ **New terms** must land in the **glossary** first – otherwise no merge.

Not documentation but an artifact to validate against – so that **change gets cheaper and safer**.



The right context makes AI far more reliable.

A model like the Mental Model gives AI the right context: it does not eliminate hallucinations – but it shrinks the space where AI has to guess, moving it from guessing to grounded reasoning. The cleaner the structure, the better the context. In practice: the Mental Model as a context layer for AI assistants.



The pendulum swings – back to spec-driven development.

From “no big design up front” to spec-first – because AI makes specifications fast and affordable.

For years upfront modeling was too expensive, so agile deliberately dropped it. That math has changed.

Tools come and go – the model stays.

The standard stays: capture essential complexity. Only now the path is open to everyone.

That is how I still build today: close to the code, but with the model in mind. Thank you.

